



Pranav Khare <khare.pranav@gmail.com>

Invitation to Review for IEEE Access Access-2024-30103

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Thu, Sep 26, 2024 at 11:57 PM

Reply-To: ieeeaccess2@ieee.org

To: khare.pranav@gmail.com

27-Sep-2024

Dear Mr. Khare:

The manuscript entitled "Tensor Factorization-Based Approach for Knowledge Graphs to Bridge Inventory Data Gaps of Manufacturing Processes" has been submitted to IEEE Access. Since your expertise is relevant to this article, we would be grateful if you would consider reviewing it.

The abstract appears at the end of this letter, along with the names of the authors. Please let me know within 2 days if you will be able to accept my invitation to review. If you agree to review, we ask that you provide your review comments within **7 days** to assist IEEE Access in maintaining its expedited peer review process of 4 to 6 weeks. If you are interested in reviewing but need additional time to complete the review, we can grant a reasonable deadline extension upon request. For additional information on IEEE Access and tips on completing a quality review, please visit our [Reviewer Best Practices guide](#).

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Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Prof. Derek Abbott
Editor-in-Chief, IEEE Access

ieeeaccess2@ieee.org

MANUSCRIPT DETAILS

TYPE: Research Article

TITLE: Tensor Factorization-Based Approach for Knowledge Graphs to Bridge Inventory Data Gaps of Manufacturing Processes

AUTHORS: Saad, Mohamed; Zhang, Yingzhong; Tian, Jinghai

ABSTRACT: Life cycle inventory gaps are a prevalent issue influencing the life cycle assessment of a product system, leading to inaccurate sustainable decision-making. This paper introduces a computational methodology to bridge LCI data gaps for manufacturing processes based on statistical relational learning for Knowledge graphs. First, an LCI data completion-oriented ontology-based model is developed, providing a formal and semantic representation of interconnected concepts of LCI data. Accordingly, a knowledge graph of LCI data for manufacturing processes (MPKG) in the Ecoinvent database has been constructed as a network of nodes connected by edges (relations). Second, the MPKG is represented mathematically into a third-order tensor that models all possible facts in the MPKG as triples. In this approach, the data gaps (flows and their values) are modeled as unknown triples given the subject node and the relation, while the object nodes representing the data gap to be estimated are missing. Then, a tensor factorization-based relational model is applied to decompose the tensor model into factor matrices that capture the semantic similarity of entities and relations in the MPKG using their latent feature representations. Finally, the data gaps are estimated via the computation of the score vector for each unknown triple, where the highest rank indicates the most likely object node in the MPKG to complete the triple, representing the missing flow or its corresponding value. Moreover, the estimated data are compared to the original data to evaluate the model performance. The results emphasize that our approach has considerable predictive potential to bridge flow data gaps for the manufacturing process.

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